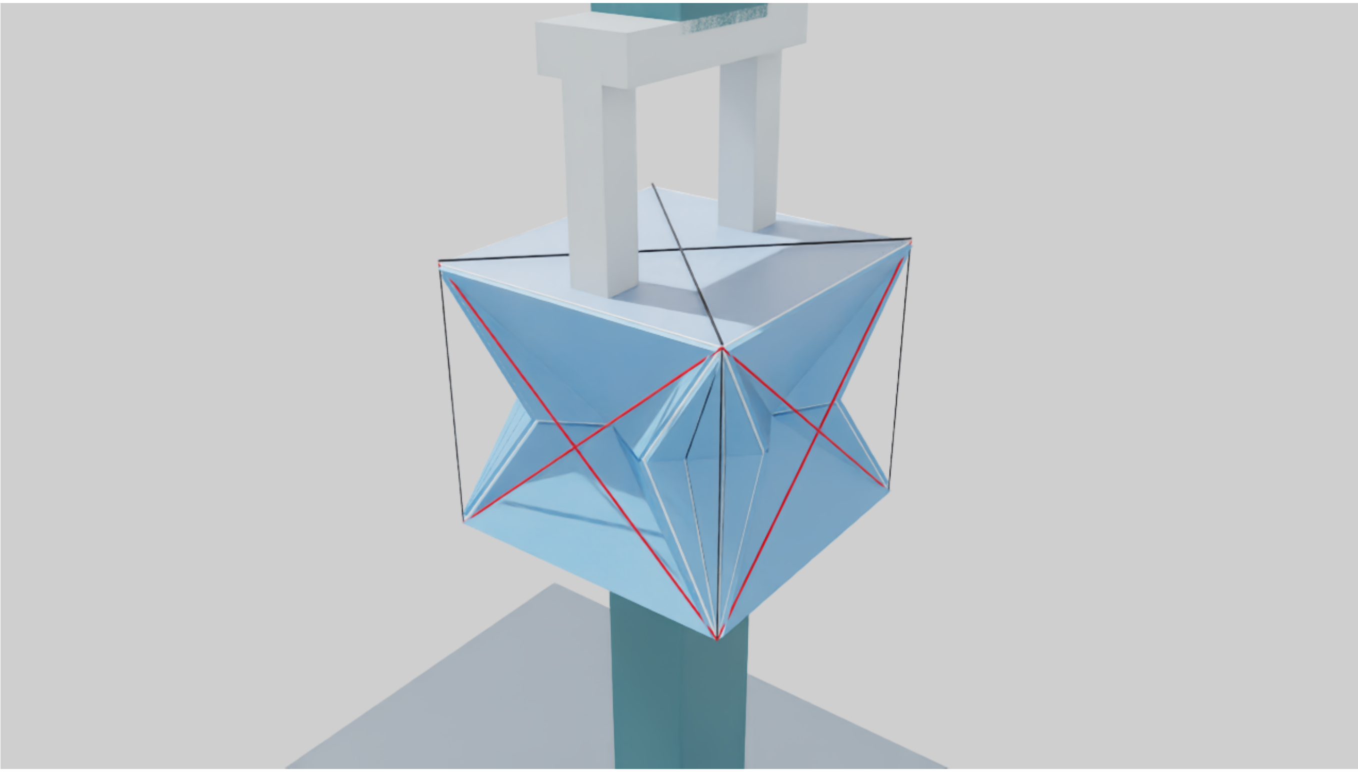
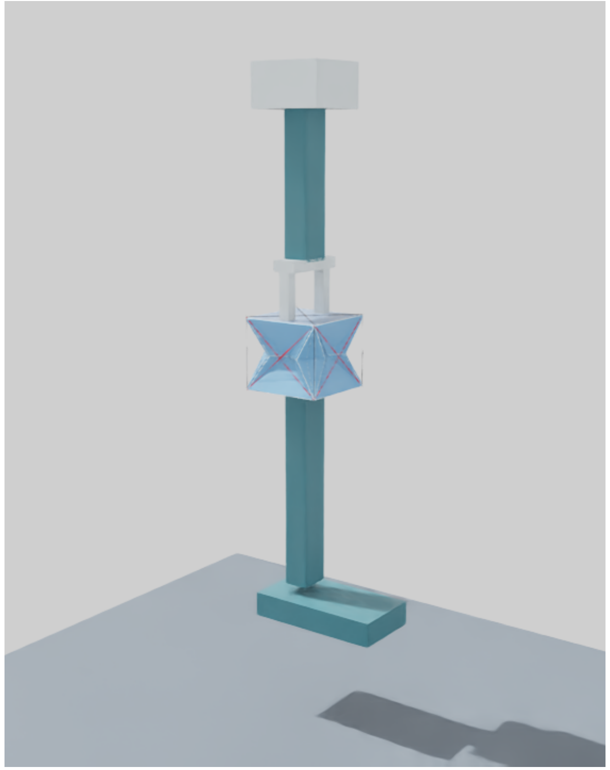
DESIGN CHECKPOINT

Your original joint. At the knee only.

Original JSON geometry, rigid end plates and cable-load-driven finite-element response inside Isaac Sim.



Original fold pattern with red cross-plate cables and black top routing reference.



One origami knee; rigid thigh, shank and foot.

EXACT GEOMETRY

28 shared vertices
50 panels / 76 edges
30 x 30 x 26.4 mm neutral surface

CONFIRMED STACK

PLA: 0.4 mm
PET film: 80 micrometres
0.2 mm exposed PET gap is provisional

CABLE INPUTS

Compression, two-plane bending, twist
Experimental nonnegative tensions in N
Both square plates remain rigid

EXPLORATORY - NOT A LARGE-FOLDING OR STRENGTH VALIDATION

The current live FEM is quasistatic and small-strain. Mesh convergence fails; no material allowables, fatigue life or motor rating are claimed.

How to demonstrate it and what to report

Use the "Exact knee - cable FEM" panel. The custom solver is controlled here, not by the Isaac timeline Play button.

01 **Inspect -** Choose Whole leg or Exact knee close-up. The displayed movement is true scale, with no hidden amplification.

02 **Apply cable load -** Select a pattern, enter tension in N per active cable and click Apply. Start at 0.1 N; 0.25 N is the demo input, not a hardware recommendation.

03 **See FEM results -** Stress / material toggles the color field. Read bend, twist, compression, PET/PLA stress and strain in the panel.

04 **Change one parameter -** Adjust PET gap, PET/PLA modulus or mesh refinement. Rebuild at zero load, then repeat the same cable load.

05 **Save and compare -** Save FEM results and inputs creates a new timestamped record. Pause freezes the solver; Neutral removes cable loads.

CHECKS AND LIMITS

7 / 7 numeric tests passed.
7 / 7 cable patterns exercised on the single knee.
No hip or ankle origami copies in the active scene.
Rigid roof distances preserved to display precision.

MESH SENSITIVITY AT 0.1 N / CABLE

Refinement	Compression	CW twist
1	2.521 um	-0.003275 deg
2	4.984 um	-0.005009 deg
3	6.207 um	-0.006089 deg

Refinement 2 to 3 changes compression by 24.6% and twist by 21.6%. Both exceed the 5% convergence criterion.

WHAT IS BEING SOLVED

Quadratic TET10 PLA/PET elements, four Gauss points, perfect bonds. Interior FEM coordinates are condensed exactly; cable forces determine the rigid lower plate pose. Displacement, strain and stress are recovered live.

Next: converge and calibrate the creases, then validate nonlinear folding, guide clearance and loaded-leg behavior. No ML policy has been trained.

Reproduction and full assumptions: exact_joint/README.md. Source JSON SHA256 begins d578618a0f90. Starting shared source checkpoint: 65e98b1c.

Element reference: FEBio Theory Manual 4.7, section 4.1.4. Model-limit reference: SOLIDWORKS Help, Large Displacement Solution.